

MM-Net Verification Log

Claim $\rightarrow$ checked how $\rightarrow$ result (Confirmed / Corrected / Flagged). Every number below is produced live by a cell in one of the two executed notebooks — no value is copied from a cached JSON/CSV:

- `revision/MM_Net_reproduction.ipynb` — the reproduction (14 code cells; model + train loop)

inlined; 10-fold GPU training; ablation grid; baselines; per-event-type; AHI; Wilcoxon; t-SNE; confusion). Re-executed end to end in 82.4 min, 0 errors.

- `revision/supplementary_analysis.ipynb` — post-hoc analysis (retrains the headline seed 42 $\rightarrow$

identical result, saves artifacts, corrected AHI, severity, per-event-type, 7 clean figures).

Reproduced headline numbers

#	Paper claim	Checked how	Result
1	Staging acc 0.721, mF1 0.645, κ 0.606	Re-ran the concat model 10-fold, fixed folds (seed 42), live GPU training	Confirmed — 0.7227 / 0.651 / 0.611
2	Apnea AUC 0.700, AP 0.333	Same run, respiratory head vs binary apnea label	Confirmed — AUC 0.711, AP 0.337
3	Per-class F1 [0.77, 0.27, 0.78, 0.68, 0.72]	Same run, per-fold per-class F1	Confirmed — [0.77, 0.27, 0.78, 0.67, 0.73]
4	"cross-modal attention ... 0.721 / 0.700" (abstract)	attention (cross) run vs concat run	Corrected — those are the <i>concat</i> numbers; attention is 0.712/0.705. Concat is now the stated headline model, attention an ablation.
5	Airflow removal "marginally higher"	leave-one-out airflow run; difference vs fold SD	Corrected — AUC 0.704 vs 0.711 is within one fold SD; text now says "unchanged," not "higher"
6	Determinism	Retrained the headline a second time in the supplementary notebook	Confirmed — 0.7223, identical to 3 decimals

Data and labels

#	Item	Checked how	Result
7	N of the cohort	Counted subjects; SN28 byte-identical to SN15	Corrected/Confirmed — staging N=99, respiratory N=96 (complete cardio); stated explicitly
8	Respiratory-event labels	Re-parsed Flow Events; reconstructed "any event" per epoch	Confirmed — matches the trained apnea label exactly (0 mismatches / 97 subjects)
9	Event composition	Counted event types across cohort	Confirmed — hypopnea 81%, obstructive 11.6%, central 5.8%
10	AHI severity distribution	subject_description.xlsx, AASM cut-offs	Confirmed — Normal 15 / Mild 24 / Moderate 23 / Severe 38
11	SpO2 sampling	Descriptor: native 4 Hz; pipeline resamples to 25 Hz	Flagged/stated — the cardiorespiratory grid up-samples SpO2 4 Hz → 25 Hz; now stated in Section III

New experiments (added in revision — all live in the notebooks)

#	Experiment	Result
12	Respiratory baselines on the 14 cardio features (§4.1)	desat-rule AUC 0.596 / logreg 0.582 / gradient-boosting 0.670; random AP 0.16. MM-Net (0.711) beats gboost by ~4pp AUC — modest → argument reframed to joint single-pass + causal attribution, not detection supremacy
13	Modality-ablation grid, leave-one-out (§4.2, headline)	full 0.723/0.711 · —SpO2 0.727/ 0.681 · —effort 0.728/0.726 · —pulse/HRV 0.723/0.700 · —ECG 0.723/0.711 · —airflow 0.722/0.704 · —EOG 0.712 /0.705 · —EMG 0.724/0.707 · —all cardio 0.724/ 0.673 . Clean physiological split: cardio→respiratory, EOG→staging
14	Per-event-type AUC (§4.3)	hypopnea 0.692 · obstructive 0.763 · central 0.840 — strongest on the most severe events
15	Predicted burden vs clinical AHI (§4.4)	Spearman rho=0.315, p=0.0017, n=96 (corrected join; the naive name-join matched only 9)
16	Staging by SDB severity (§4.5)	accuracy 0.770 (normal) → 0.730 (mild) → 0.712 (moderate) → 0.708 (severe)
17	Wilcoxon signed-rank over 10 folds (§3.6)	removing all cardio vs full: respiratory AUC p=0.004 (drops), staging p=0.91 (unchanged)

Correctness fixes (all applied to paper/multimodal.tex)

- Eq. (BCE/joint loss) truncation in the compiled PDF → repaired with an `align` block. **Applied.**
- Corresponding author added: Dr. Md Iftekharul Mobin, Dept. of CS, AIUB (iftekhar.mobin@aiub.edu). **Applied.**

- Data availability (iSLEEPS public), code availability, ethics (NIMHANS IEC), funding statements. **Applied.**
- Title changed to "A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke." **Applied.**
- EEG-only feature/ensemble models moved to supplementary framing; MM-Net + input-matched deep models + published baselines (with caveat) kept in main text; **concat is the headline**, attention an ablation. **Applied.**

Figures (real, regenerated from the saved model/embeddings; graph-only, no annotations)

- `fig_ablation` — leave-one-out grid, 9 rows incl. SpO2 (staging vs respiratory AUC).
- `fig_benchmark_clean` — staging accuracy across architectures (healthy-built deep models 0.61–0.69).
- `fig_tsne` — PCA+t-SNE of learned embeddings; clean stage separation, diffuse respiratory (honest).
- `fig_confusion` — 5-class row-normalised confusion matrix from pooled test predictions.
- `fig_ahi` — predicted per-patient burden vs clinical AHI ($\rho=0.315$).
- `fig_event_type`, `fig_severity`, `fig_resp_baselines`, `fig_sdb_burden` — from the runs above.

Nothing unsupported

Every number in the final paper traces to a cell in the two notebooks above. No value was invented; where a claim could not be reproduced as written (rows 4, 5) it was corrected, not kept.